ENGINEERING PROJECT DOCUMENTATION

LANDSLIDE EARLY WARNING SYSTEM

MATLAB–Simulink basedrainfall-driven slope-instability simulation with multi-parameter sensor fusion and estimated Time-to-Failure

CASE STUDY

Munnar–Idukki region, Western Ghats, Kerala, India

PROJECT TYPE

Simulation-based proof of concept

PLATFORM

MATLAB and Simulink

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01 • INTRODUCTION

Document overview

This document describes a MATLAB–Simulink proof-of-concept for a Landslide Early Warning System (LEWS). The simulation represents rainfall-driven slope wetting and instability, combines hydrological, stability and deformation indicators, and presents staged warning states together with an estimated remaining Time-to-Failure (TTF).

The selected case-study context is the Munnar–Idukki region of the Western Ghats in Kerala, India. The model is intended to demonstrate an integrated monitoring and warning workflow; it is not a field-validated operational warning device.

Important scope statement: All outputs described here are simulation-based. The model does not establish real-world warning accuracy, site-specific safety thresholds, or operational readiness.

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02 • PROJECT DEFINITION

Project background and objectives

Landslides can develop as rainfall infiltrates a slope, increases soil moisture and pore-water pressure, and changes the forces acting within the soil. Monitoring rainfall alone may not fully represent the evolving condition of a slope. This project therefore brings multiple indicators into a single simulation and warning workflow.

The system links conceptual hydrological and geotechnical responses with a multi-input risk assessment, a four-stage warning output and a forward-looking TTF estimate.

Project objectives

01. Represent rainfall-driven soil wetting and the associated evolution of pore-water pressure.

02. Represent slope stability using a Factor of Safety (FS) indicator.

03. Track surface and subsurface deformation using tilt and shear-strain indicators.

04. Combine six model inputs into a risk-oriented warning workflow.

05. Display Green, Yellow, Orange and Red warning states.

06. Estimate remaining Time-to-Failure (TTF) as a simulation output.

Intended use

The project is suitable for demonstrating the structure of an early-warning workflow, exploring how several indicators may evolve together, and supporting further development of a sensor-fusion model. It should not be used to make real evacuation, engineering or public-safety decisions.

03 • MODEL BASIS

Physical concept and monitored parameters

The simulation follows a conceptual rainfall-to-instability progression. Rainfall is applied as an external input. As infiltration and wetting progress, soil moisture and pore-water pressure evolve. The stability indicator may decline as the slope weakens, while tilt and subsurface shear strain represent deformation-related behaviour.



Monitored inputs and indicators

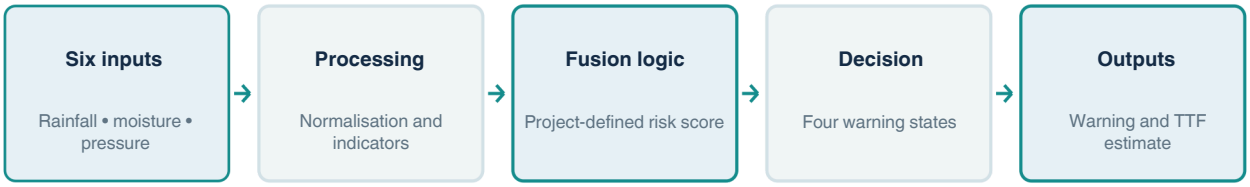
Parameter	Unit / representation	Role in the model
Rainfall intensity	mm/h	Represents the rainfall forcing applied to the slope.
Soil moisture	%	Tracks the wetting condition of the soil.
Pore-water pressure	kPa	Represents subsurface pressure response during wetting.
Factor of Safety (FS)	Dimensionless	Represents the modelled slope-stability condition; a declining value indicates reduced stability.
Surface tilt	Degrees (°)	Represents surface movement or rotation.
Subsurface shear strain	με	Represents deformation within the subsurface soil.

Simulation time scaling: 1 simulation second represents 0.6 physical hours. This fixed scaling accelerates the simulated progression for demonstration purposes; it is not a measured field-time relationship.

04 • SYSTEM DESIGN

System architecture and signal flow

The architecture is organised as a staged processing chain. Six monitored or model-derived inputs are brought into the processing stage, represented on a common scale, combined through project-defined fusion logic, and used to produce warning and TTF outputs.



Input group

Input category	Signals
Hydrological	Rainfall intensity, soil moisture, pore-water pressure
Stability	Factor of Safety
Deformation	Surface tilt, subsurface shear strain

Output group

Output	Description
Risk / severity indicator	A combined indicator used by the project-defined warning logic.
Warning state	One of four displayed states: Green, Yellow, Orange or Red.
Estimated TTF	A forward-looking estimate of remaining time to failure, displayed in hours.

Implementation note

The six-input fusion stage and warning logic are project implementation elements. The design is a simulation workflow, not evidence of a deployed or calibrated field sensor network. Exact thresholds and weights should be documented from the implemented Simulink blocks before being treated as fixed engineering parameters.

05 • PROCESSING WORKFLOW

Methodology and AI sensor fusion

1. Input generation

Generateor providetime-varying rainfall, soil moisture, pore-water pressure, Factor of Safety, tilt and shear-strain signals.

2. Signal preparation

Bringinputsignalsinto a formsuitable for combined processing. The project documentation describes normalisation as part of the processing stage.

3. Physical-response representation

Representrainfall-drivenwetting,pore-pressureresponse, stability change and deformation progression using the MATLAB–Simulink model.

4. Risk fusion

Combinethesixindicators through project-defined fusion logic to obtain a risk/severity indicator.

5. Warning-state decision

Maptheevolvingriskconditiontooneof the four warning states: Green, Yellow, Orange or Red.

6. TTF estimation

Usethemodel'sregression-based stage to estimate remaining time to failure and display the estimate in hours.

AI / regression component

Theproject describes aregression-based TTF estimate driven by the evolving input indicators. A reliable evaluation requires a clearly documented dataset, an appropriate train/test separation, and testing on unseen events. The supplied documentation does not provide enough clearly readable information to verify the model's full training procedure or independently validate its reported performance.

Interpretation: In this proof of concept, “AI sensor fusion” refers to the project’s combined-indicator and regression workflow. It should not be described as a validated predictive model until its training data, evaluation method and independent test results are established.

06 • CASE-STUDY DATA

Simulation setup and available data

The source documentation provides two clearly identifiable reference states. The 0-hour state is the initial stable condition. A second state is reported at 18 hours, with increased rainfall and soil moisture and a lower Factor of Safety. The table below intentionally limits itself to those legible reference values.

Variable	At 0 h	At 18 h
Estimated TTF	42.0 h	24.0 h
Rainfall intensity	0 mm/h	5 mm/h
Cumulative rainfall	Not stated	85 mm
Soil moisture	18.0%	38.2%
Pore-water pressure	−15 kPa	Not clearly specified
Factor of Safety	1.85	Not clearly specified in the legible reference row
Surface tilt	0.02°	Not clearly specified in the legible reference row
Shear strain	12 με	Not clearly specified in the legible reference row
Warning state	Green	Not clearly specified

Data quality note

The uploaded source PDF contains a longer intermediate-results table, but several entries are merged or misaligned in the extracted document. Those ambiguous values have not been reproduced as verified measurements in this report. Replace or extend this table using the original MATLAB/Simulink output or source dataset if additional time points are required.

Suggested figures to add from the simulation

For a complete technical submission, export the actual Simulink plots for rainfall and cumulative rainfall, soil moisture, pore-water pressure, Factor of Safety, tilt, shear strain, warning-state progression and TTF versus elapsed time. Each plot should include axis labels, units, a descriptive caption and an indication that the data are simulated.

Results, interpretation and limitations

The two clearly legible reference states show the intended direction of the simulation: by 18 hours, the reported rainfall intensity has reached 5 mm/h, cumulative rainfall is 85 mm, soil moisture has increased from 18.0% to 38.2%, and estimated TTF has decreased from 42.0 hours to 24.0 hours.

These values illustrate the behaviour encoded in the simulation. They do not establish a real slope's probability of failure or prove that the estimated TTF would be accurate during an actual landslide event.

Key interpretation points

• Rainfall is treated as the external forcing variable.

• Rising soil moisture represents progressive wetting in the simulated slope.

• Pore-water pressure and Factor of Safety are used to represent subsurface response and stability condition. • Tilt and shear strain provide deformation-related indicators.

• The warning state and TTF are model outputs that depend on the implemented fusion and regression logic.

Limitations

• The model is a proof of concept and is not field-validated.

• Input signals, thresholds and response relationships require site-specific calibration.

• The supplied document does not establish that the six inputs originate from a deployed sensor network.

• Regression performance cannot be independently confirmed from the readable information supplied.

• Results from one simulated case-study sequence do not demonstrate generalisation to other slopes, rainfall events or geological conditions.

• An operational system would require reliable instrumentation, communications, data-quality checks, fail-safe alert handling and validation by qualified geotechnical specialists.

Safety disclaimer: This simulation is for academic demonstration and research development only. It must not be used as a substitute for official landslide warnings, field monitoring, geotechnical assessment or emergency-management procedures.

08 • REPRODUCIBILITY

Implementation files and future development

The uploaded project documentation identifies the following MATLAB/Simulink deliverables. Confirm that these files exist in the final project folder and that their names match the actual files before publishing the repository.

File / deliverable	Purpose
matlab_landslide.slx	Simulink model file identified in the project documentation.
Matrix_project_LEWS.m	MATLAB script identified in the project documentation.

Reproduction checklist

1. Place the MATLAB script and Simulink model in the same project folder, or update the script paths.

2. Open the project using a compatible MATLAB/Simulink release.

3. Run the MATLAB script and open the Simulink model as required by the implementation.

4. Check that all six input signals are available and have consistent time bases and units.

5. Run the simulation and inspect the warning-state and TTF outputs.

6. Export plots and record the exact model settings, thresholds and dataset used.

Recommended future work

• Document the exact risk-score equation, input normalisation and warning thresholds from the implemented blocks.

• Use a larger, traceable dataset containing multiple rainfall events and slope conditions.

• Separate training, validation and test data by event to reduce data leakage.

• Evaluate regression performance on unseen events and report MAE, RMSE and R^2 with the test-set definition.

• Calibrate the physical response and warning thresholds using instrumented, site-specific field data.

• Add real sensor acquisition, communications, data-quality monitoring and alert delivery.

• Conduct independent geotechnical review and field validation before any operational use.

Conclusion

The Landslide Early Warning System project demonstrates a structured MATLAB–Simulink workflow that connects rainfall-driven slope response, six monitored indicators, a staged warning output and an estimated Time-to-Failure. The next major step is to strengthen reproducibility and validate the physical and predictive components using traceable data and independent field evidence.

End of document • Landslide Early Warning System • 2026